

2.1 The Periodic Table: Chemical Periodicity

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.1 The Periodic Table: Chemical Periodicity
Difficulty	Easy

Time allowed: 40

Score: /29

Percentage: /100

Question 1a

The elements across Period 3 show patterns in their chemical and physical properties.

State the trend in the atomic radius of successive elements in Period 3.

[1 mark]

Question 1b

The electrical conductivity of the elements in Period 3 varies across the period as shown in Fig. 1.1.

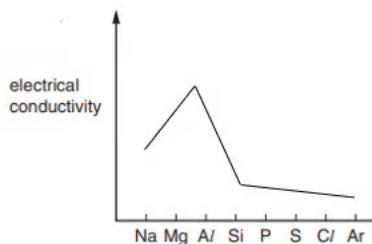


Fig. 1.1

i)

Explain why aluminium has a much higher electrical conductivity than sulfur. You should refer to the structure of both elements.

[2]

ii)

Explain why aluminium has a higher electrical conductivity than magnesium.

[2]

[4 marks]

Question 1c

The melting points of three successive elements in Period 3 are shown in Table 1.1.

Table 1.1

Element	Melting point / °C
Aluminium	660
Silicon	1410
Phosphorous	44

i)

Explain, referring to its structure and bonding, why the melting point of silicon is high.

[3]

ii)

Explain, referring to its structure and bonding, why the melting point of phosphorus is low.

[3]

[6 marks]

Question 2a

The melting points of some of the oxides of the Period 3 elements, sodium to sulfur, are given in Table 2.1

Table 2.1

Compound	Melting point / K
Na ₂ O	1193
MgO	3173
Al ₂ O ₃	2313
SiO ₂	1883
P ₄ O ₁₀	613
SO ₂	198

State what is broken when each of the following compounds is melted.

Na₂O

SiO₂

P₄O₁₀

[3 marks]

Question 2b

i)

Give the formula of a Period 3 oxide from Table 2.1 that reacts with water to produce a hydroxide.

[1]

ii)

Give the formula of a Period 3 oxide from Table 2.1 that reacts with water to produce an acid.

[1]

iii)

Give the formula of a Period 3 oxide from Table 2.1 that has no reaction at all with water.

[1]

[3 marks]

Question 2c

The oxides of the elements of the third Period behave differently with NaOH (aq) and HCl (aq). In some cases, no reaction occurs.

Complete Table 2.2 by writing 'reaction' for any reaction that occurs between the Period 3 oxide and HCl (aq) or NaOH (aq), with heating if necessary. If you think no reaction takes place write 'no reaction'.

Table 2.2

Period 3 oxide	With HCl (aq)	With NaOH (aq)
MgO (s)		
Al ₂ O ₃ (s)		
SO ₂ (g)		

[3 marks]

Question 3a

Period 3 elements react with chlorine to form chlorides.

i)

Give the equation to show the reaction when aluminium is heated with chlorine to form aluminium chloride, Al_2Cl_6 .

[1]

ii)

Complete Table 3.1 to show the oxidation number of the Period 3 elements in the chloride with the formula given in the table.

Table 3.1

Period 3 element	Formula of chloride	Oxidation number of Period 3 element in the chloride
Na	NaCl	
Mg	MgCl_2	
Al	Al_2Cl_6	
Si	SiCl_4	
P	PCl_5	
S	SCl_2	

[3]

[4 marks]

Question 3b

i)

State what would be seen when MgCl_2 is added to water.

[1]

ii)

Suggest the pH of the resulting solution.

[1]

[2 marks]**Question 3c**

Liquid silicon chloride, SiCl_4 , and solid phosphorus(V) chloride, PCl_5 , can be hydrolysed in water, both reactions produce acidic solutions and release white fumes.

Give the balanced symbol equation for the reaction of phosphorus(V) chloride with water. Include state symbols.

[3 marks]

Model Answers: Easy

1a

a) The trend in the atomic radius of successive elements in Period 3 is:

- Decreases (across the period); [1 mark]

[Total: 1 mark]

- This can be counter-intuitive and students often give the incorrect answer
- As you across a period, the nuclear charge increases as each successive element has an extra proton
- The attraction between the outer electrons and the nucleus is therefore greater, which pulls the outer electrons closer to the nucleus, decreasing the atomic radius

1b

b)

i) Aluminium has a much higher electrical conductivity than sulfur because:

- Aluminium has a giant metallic lattice / structure

AND

With free / delocalised electrons (to carry the charge); [1 mark]

- Sulfur is a simple molecule / has a simple molecular structure

AND

With no free / delocalised electrons (to carry the charge); [1 mark]

ii) Aluminium has a higher electrical conductivity than magnesium because:

- Aluminium has more outer electrons (than magnesium)

OR

Aluminum has 3 outer electrons **AND** magnesium has 2 outer electrons; [1 mark]

- So it has more delocalised electrons (than magnesium) to carry the charge; [1 mark]

[Total: 4 marks]

- When told to refer to the structure of elements, you must identify the type of structure that each element exists as
- As these are all elements, the bonding will either be metallic or covalent
 - Ionic bonding only exists between ions of different elements
- This means that when you refer to whether an element conducts electricity, you need to refer to whether the structure contains free / delocalised electrons that are able to carry the charge

1c

c)

i) The melting point of silicon is high because:

- Silicon has a giant molecular lattice / structure; [1 mark]
- With strong covalent bonds (between silicon atoms); [1 mark]
- Which require a large amount of energy to overcome; [1 mark]

ii) The melting point of phosphorus is low because:

- Phosphorus has a simple molecular structure / is a simple molecule; [1 mark]
- With weak instantaneous dipole-induced dipole forces / id-id forces / intermolecular forces; [1 mark]
- Which only require a small amount of energy to overcome; [1 mark]

[Total: 6 marks]

- For each explanation, you need to state:
 - The type of structure that the element has
 - The type and strength of bonds / forces between atoms / molecules
 - The relative amount of energy needed to overcome these bonds / forces

2a

a) The bond / force that is broken when each of the following compounds is melted is:

- Na_2O : ionic; [1 mark]
- SiO_2 : covalent; [1 mark]
- P_4O_{10} : van der Waals' forces / intermolecular forces / induced dipoles; [1 mark]

[Total: 3 marks]

- Na_2O is a giant ionic structure where all the oppositely charged ions experience a strong force of electrostatic attraction, which is ionic bonding
 - When it melts, it is the strong forces of attraction / ionic bonds between oppositely charged ions that are overcome
- SiO_2 is a giant molecular structure where every silicon atom is covalently bonded to 4 oxygen atoms and each oxygen atom is covalently bonded to 2 silicon atoms
 - When it melts, it is these strong covalent bonds that are overcome
- P_4O_{10} has a simple molecular structure with strong covalent bonds between atoms within a molecule but weak intermolecular forces of attraction
 - When it melts, it is the weak intermolecular forces of attraction that are overcome

2b

b)

i) The formula of a Period 3 oxide from Table 2.1 that reacts with water to form a hydroxide is:

- Na_2O
OR
 MgO ; [1 mark]

ii) The formula of a Period 3 oxide from Table 2.1 that reacts with water to produce an acid is:

- P_4O_{10}
OR
 SO_2 ; [1 mark]

iii) The formula of a Period 3 oxide from Table 2.1 that has no reaction at all with water is:

- Al_2O_3
OR
 SiO_2 ; [1 mark]

[Total: 3 marks]

- Na_2O and MgO reacts with water to produce an aqueous solution of the metal

hydroxide

- P_4O_{10} and SO_2 react with water, to produce phosphoric acid and sulfuric acid respectively
- Al_2O_3 or SiO_2 do not dissolve or react in water
- The pH of the resulting solutions of those Period 3 oxides that do react or dissolve in water decreases (becomes more acidic) as you go across the period

2c

c) The completed table to show where a reaction occurs between Period 3 oxides and HCl (aq) and NaOH (aq) is:

Period 3 oxide	With HCl (aq)	With NaOH (aq)
MgO (s)	reaction	no reaction
Al_2O_3 (s)	reaction	reaction
SO_2 (g)	no reaction	reaction

- Each correct row; [1 mark]

[Total: 3 marks]

- MgO is a basic oxide so will react with acids, but not bases
- Al_2O_3 is an amphoteric oxide so will react with both acids and bases
- SO_2 is an acidic oxide so will react with bases but not acids
- The reactions that will occur are:
 - $MgO + 2HCl \rightarrow MgCl_2 + H_2O$
 - $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$
 - $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

OR

 - $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
 - $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

3a

a)

i) The equation to show the reaction of aluminium with chlorine to form aluminium chloride, Al_2Cl_6 is:

- $2\text{Al} + 3\text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$; [1 mark]

The completed table to show the oxidation numbers of the Period 3 elements in their chlorides is:

Period 3 element	Formula of chloride	Oxidation number of Period 3 element in the chloride
Na	NaCl	+1
Mg	MgCl ₂	+2
Al	Al ₂ Cl ₆	+3
Si	SiCl ₄	+4
P	PCl ₅	+5
S	SCl ₂	+2

- All 6 oxidation numbers correct; [3 marks]
- 4 or 5 oxidation numbers correct; [2 marks]
- 2 or 3 oxidation numbers correct; [1 mark]
- 1 oxidation number correct; [0 marks]

[Total: 4 marks]

- You are told the formula of aluminium chloride so you can construct the unbalanced equation:
 - $\text{Al} + \text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$
- Balance the Al atoms:
 - $2\text{Al} + \text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$
- Balance the Cl atoms:
 - $2\text{Al} + 3\text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$
- **Remember:** The oxidation number of the compound is 0
- Chlorine is more electronegative than the other Period 3 elements so will have a negative oxidation number (-1)
- The Period 3 element that the chlorine is bonded to will have a positive oxidation number to cancel out the charge and give an electronically neutral compound
- The elements from Na to P use all of their outer shell electrons in bonding

	chloride	element in the chloride
Na	NaCl	+1
Mg	MgCl ₂	+2
Al	Al ₂ Cl ₆	+3
Si	SiCl ₄	+4
P	PCl ₅	+5
S	SCl ₂	+2

- All 6 oxidation numbers correct; [3 marks]
- 4 or 5 oxidation numbers correct; [2 marks]
- 2 or 3 oxidation numbers correct; [1 mark]
- 1 oxidation number correct; [0 marks]

[Total: 4 marks]

- You are told the formula of aluminium chloride so you can construct the unbalanced equation:
 - $\text{Al} + \text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$
- Balance the Al atoms:
 - $2\text{Al} + \text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$
- Balance the Cl atoms:
 - $2\text{Al} + 3\text{Cl}_2 \rightarrow \text{Al}_2\text{Cl}_6$
- **Remember:** The oxidation number of the compound is 0
- Chlorine is more electronegative than the other Period 3 elements so will have a negative oxidation number (-1)
- The Period 3 element that the chlorine is bonded to will have a positive oxidation number to cancel out the charge and give an electronically neutral compound
- The elements from Na to P use all of their outer shell electrons in bonding

3b

b)

i) When MgCl₂ is added to water, you would see:

- MgCl₂ / white solid disappears / dissolves; [1 mark]
- To form a colourless solution; [1 mark]

ii) The pH of the resulting solution is:

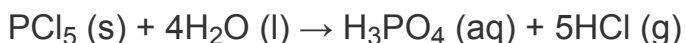
- 6.5; [1 mark]

[Total: 3 marks]

- You need to be able to write equations for the reactions of the Period 3 chlorides from Na to P with water and be able to predict the likely pH of the resulting solutions

3c

c) The balanced symbol equation for the reaction of phosphorus(V) chloride with water, including state symbols is:



- Correct formulae; [1 mark]
- Correct balancing; [1 mark]
- Correct state symbols; [1 mark]

[Total: 3 marks]

- You are given the formula of phosphorus(V) chloride and told that it is a solid
- You know the reaction is with water so you can write the reactants:
 - $\text{PCl}_5 (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightarrow$
- You should know that a solution of phosphoric acid is produced
- You are told that white fumes are produced which should indicate that HCl gas is evolved, giving the unbalanced equation:
 - $\text{PCl}_5 (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_3\text{PO}_4 (\text{aq}) + \text{HCl} (\text{g})$
- Balance the Cl atoms by putting a 5 before HCl on the right-hand side:
 - $\text{PCl}_5 (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_3\text{PO}_4 (\text{aq}) + 5\text{HCl} (\text{g})$
- This gives 8 H atoms on the right-hand side which can be balanced by putting a 4 before H₂O on the left-hand side:
 - $\text{PCl}_5 (\text{s}) + 4\text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_3\text{PO}_4 (\text{aq}) + 5\text{HCl} (\text{g})$